Comprehensive Guide: UX Research and Statistics

1. Usability Testing Fundamentals and Metrics

💡 Core Definitions

- **UI - design:** How the product looks like.
- **UX - user experience:** The 'feel' of the product.
- **Usability:** The extent to which the user ensures **effectiveness**, **efficiency**, and **satisfaction** of a product. It is the measure of the product's quality.
- **Usability Testing:** The practice of testing how easy a design is to use with a group of representative users.
- **Utility + Usability = Useful**.

🎯 Goals of Usability Testing

Usability testing aims to measure the following in users:

- Performance.
- Accuracy.
- Recall.
- Emotional response.

📊 Usability Metrics

These metrics help quantify the user experience:

Metric	Measurement Focus	Deals with...
🚀 **Effectiveness**	Task Completion (Qualitative)	Task completion time.
⏱️ **Efficiency**	Qualitative	Overall relative efficiency.
🧠 **Memorability**	Recall	How users remember how to carry out tasks (e.g., tested by a distraction video).

✨ **Satisfaction**	User Feedback	Subjective ratings (e.g., Likert scale).
🚫 **Error Tolerance**	Errors	Number of errors / Time.
👷 **Learnability**	Time to Proficiency	How long it takes, or how much time users are prepared to spend learning a system.

How to Count Learnability? By observing how any efficiency metric changes over time. Data is collected at **multiple times** (trials) and analyzed quantitatively.

🗠 Process of Usability Testing

1. Plan the test.
2. Recruit participants.
3. Prepare materials.
4. Setup Environment.
5. Conduct test.
6. Analyze data.
7. Report results.

2. Research Design and Methods

🔬 Types of Research

- **Descriptive research:** Observing & drawing conclusions about what is happening (X is happening).
- **Relational research:** Connection between two or more variables (X is related to Y).
    - **Relational Investigation** carries more weight than descriptive studies.
- **Experimental research:** Effect of one or more variables on one or more outcome variables (X is responsible for Y).

🔗 Variables

- **Independent (X):** Things controlled and manipulated by experimenter.

- **Dependent (Y):** Measured by experimenter (e.g., success rates, errors, completion time, satisfaction).

🔬 Experimental Design Comparison

- **30 users** are statistically valid for usability results.

Design Type	Description	Rationale
🧍 **Between Group Subject**	Each participant is only exposed to one experimental condition.	Easier logistics for simple tasks.
🔁 **Within Group**	Each participant is exposed to multiple experimental conditions.	Appropriate when investigating tasks with large individual differences, as variability is reduced (each participant serves as their own control).

💡 Methods for Testing Usability

Factor	Categories	Details
Data Type	Qualitative / Quantitative	Qual: Observational Findings. Quant: Done over internet.
Location	Remote / In-person	Remote requires physical presence; In-person does not.

Moderator	Moderated / Unmoderated	Moderated: Can be remote (Phone/Video Interviews) or in-person. Unmoderated: Without direct supervision (Session recordings).

3. Data Types and Statistics

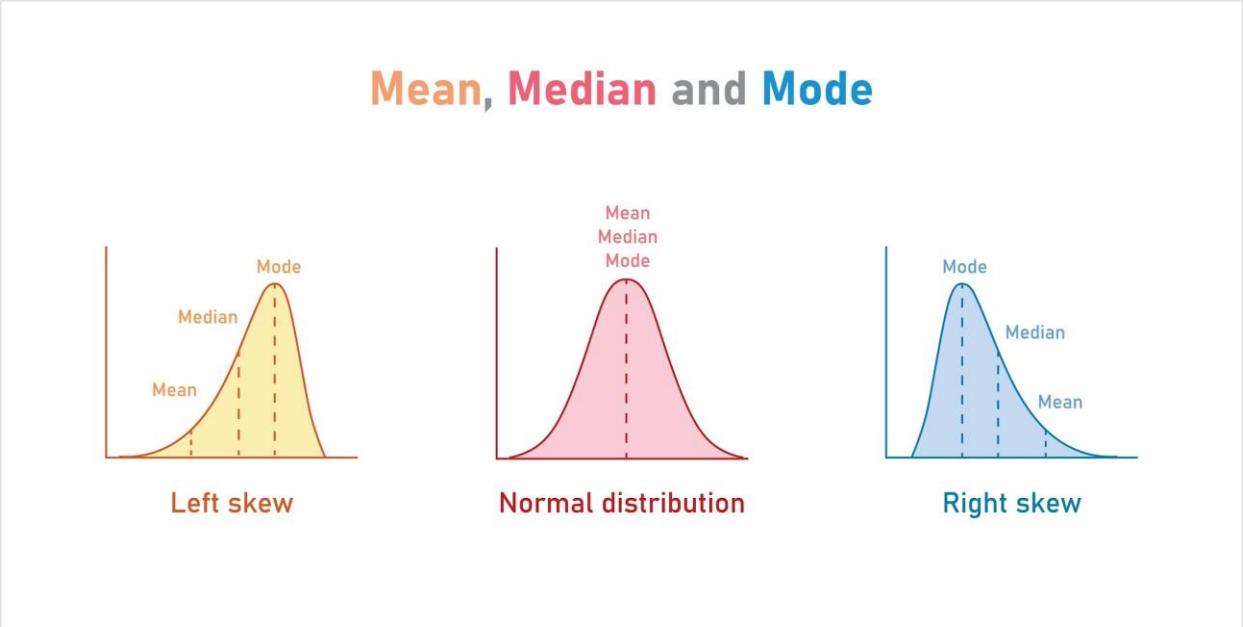
🔢 Statistical Variable Hierarchy

Statistical methods can only be used with certain data types.

- **Qualitative (Non-numeric qualities):**
  - **Nominal (unordered):** e.g., Gender.
  - **Ordinal:** e.g., Study level. (Same as Nominal but allows for **Percentiles**).
- **Quantitative:**
  - **Interval:** e.g., Scale values. **No absolute zero**.
    - e.g., User satisfaction Rating on a Likert scale, Task Difficulty Rating, Confidence Rating.
  - **Ratio:** e.g., Time Score. **Takes value of absolute zero**.

📈 Descriptive Statistics

- **Central Tendency:** Where the bulk of data is stored / located.
  - Typical Statistical Analysis $\implies$ **Mode / Mean / Median**. (Median is the **Second Quartile**).
- **Measures of Variability:** Range, Standard Deviation, Variance.
  - **Range:** distance between highest & lowest scores.
  - Population of Standard Deviation (σ):
    $$\sigma = \sqrt{\frac{\sum(X-\mu)^2}{N}} \text{}$$
    - μ - population mean.
    - N - No. of values in population.
- **Normal Distributions:** Mainly defined by 2 parameters: **Mean** and **Standard Deviation**.



📊 Data Visualization

- **Nominal data** is collected through frequencies, proportion, percentage. Visualization: **Pie chart, Bar chart**.
- **For Interval / Ratio:** $\implies$ **Box Plot / Histogram**.

4. Hypotheses and Research Questions

❓ Research Questions

1. What type of Research Question is this? **Quantitative or Qualitative**.
2. What type of Behavioural Research? **Descriptive / Relational / Mixed**.
3. Why? (For both 1 & 2 above).

💡 Research Hypotheses

- **Research Hypothesis:** A statement of expectation or prediction that will be tested by research.

- **Null-Hypothesis (H_0):** A default or no-effect statement, indicating no change / no association between variables (e.g., $\text{H}_0: (\mu_1 = \mu_2)$).
- **Alternative-Hypothesis (H_a):** Statement that there is an effect, difference or association (e.g., $\text{H}_a: (\mu_1 \ne \mu_2)$).

🔬 Types of Hypothesis

1. **Difference Hypothesis:** If there is a significant difference between 2 or more groups.
2. **Correlation Hypothesis:** Is there a relationship between at least 2 variables?
3. Undirected Hypothesis.
4. Directed Hypothesis.

- **p-Value:** Statistical measurement used to **validate a hypothesis against observed data**.

5. Ethical Guidelines and Documentation

🟢 Do's during Usability Testing

- Use **think-aloud protocol**.
- Ask more **open-ended questions**.
- Watch participant **body language**.
- Ask **follow-up questions**.
- **Record answers on video**.

🔴 Don'ts during Usability Testing

- Ask **leading questions**.
- **Judge the participants**.
- Provide **too many details**.
- Provide **too much guidance**.

🌓 Ethical Considerations

- Informed Consent.
- Privacy & Confidentiality.
- Data Security.
- Honesty.

📝 Participant Information Sheet

- Test procedure.
- What is expected.
- Role of any other people participating.
- What data is collected and how it is used?